The optimal logarithmic interpolation bound

MathDB 372857: solution of Schmidlin's Conjecture 36

Principia Math In Progress

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Task 1. Prove the conjectured interpolation bound

With the source's definitions, $x \geq 0$, $0 \leq \alpha \leq 1$, and

$$I(x,\alpha) = \widehat{\imath}(x,\alpha) = \log \frac{\Gamma(x+1)(x+1)^{\alpha}}{\Gamma(x+\alpha+1)}.$$

The identity with the integral formulation is Theorem 6 of [1], p. 6. The domain is defined on pp. 3–4 and Conjecture 36 is on p. 26.

Theorem 1. *For every $x \geq 0$ and $\alpha \in [0,1]$,*

$$I(x,\alpha) < \frac{1}{8(x+\alpha)+4}. \tag{1}$$

Moreover, 4 is the largest possible additive constant in a bound of the form $I(x,\alpha) \leq [8(x+\alpha)+C]^{-1}$ on the full domain with a positive denominator.

Lemma 1. *For $u \geq 1$, put*

$$c(u) = \log\Gamma(u) + \tfrac{1}{2}\log u - \log\Gamma(u+\tfrac{1}{2}), \qquad s = \frac{1}{(2u+1)^2}.$$

Then $c(u) < H(u)$, where

$$H(u) = \frac{1}{8(u+1)} - \frac{1}{192(u+1)^3} + \frac{1}{630(u+1)^5} + \frac{s}{2} + \frac{s^2}{4} + \frac{s^3}{6(1-s)}. \tag{2}$$

Also $c(u) = 1/(8u) - 1/(192u^3) + O(u^{-5})$ as $u \to \infty$.

Proof. For the Stirling remainder

$$r(z) = \log\Gamma(z) - (z-\tfrac{1}{2})\log z + z - \tfrac{1}{2}\log(2\pi),$$

Binet's integral [2, Eq. 5.9.10] gives, for every $z > 0$,

$$\frac{1}{12z} - \frac{1}{360z^3} < r(z) < \frac{1}{12z} - \frac{1}{360z^3} + \frac{1}{1260z^5}. \tag{3}$$

Indeed, integrate $t - t^3/3 < \arctan t < t - t^3/3 + t^5/5$ against the positive Binet kernel; these inequalities hold for all $t > 0$ by integrating the corresponding finite geometric identities for $1/(1+t^2)$. The moments are $\int_0^\infty t^j/(e^{2\pi t}-1)\,dt = j!\zeta(j+1)/(2\pi)^{j+1}$.

The Gamma duplication formula gives $c(z) = 2r(z) - r(2z)$. Applying opposite bounds from (3) therefore yields

$$c(z) < \frac{1}{8z} - \frac{1}{192z^3} + \frac{1}{630z^5},$$

and also gives the stated asymptotic expansion. The Gamma recurrence gives

$$c(u) - c(u+1) = \log \frac{u+1/2}{\sqrt{u(u+1)}} = -\tfrac{1}{2}\log(1-s).$$

Finally, for $0 < s < 1$, the positive logarithmic series implies

$$-\tfrac{1}{2}\log(1-s) \le \frac{s}{2} + \frac{s^2}{4} + \frac{s^3}{6(1-s)}.$$

Using the preceding bound for $c(u+1)$ proves (2). □

Lemma 2. *Let $\psi = \Gamma'/\Gamma$. For every $w > 0$,*

$$\frac{1}{24w^2} - \frac{7}{960w^4} \le \psi(w+\tfrac{1}{2}) - \log w \le \frac{1}{24w^2}, \tag{4}$$

$$\psi'(w+\tfrac{1}{2}) \ge \frac{1}{w} - \frac{1}{12w^3}. \tag{5}$$

Proof. We first prove the global elementary bounds

$$\frac{1}{t} - \frac{t}{6} \le \frac{1}{\sinh t} \le \frac{1}{t} - \frac{t}{6} + \frac{7t^3}{360} \qquad (t > 0). \tag{6}$$

For the left bound, expand $t - (1 - t^2/6)\sinh t$ in powers of t. The coefficient of t^{2n+1}, $n \ge 1$, is

$$\frac{(2n)(2n+1)/6 - 1}{(2n+1)!} \ge 0.$$

For the right bound, the coefficients of degrees $1, 3, 5$ in $(1 - t^2/6 + 7t^4/360)\sinh t - t$ vanish, and for $n \ge 3$ the coefficient of t^{2n+1} is

$$\frac{1 + A(7B/360 - 1/6)}{(2n+1)!} > 0, \qquad A = (2n)(2n+1), \quad B = (2n-2)(2n-1) \ge 20.$$

Both series converge everywhere, so multiplication by the positive $t\sinh t$ proves (6).

The digamma integral [2, Eq. 5.9.13], combined with Frullani's integral for $\log((w+1/2)/w)$, gives

$$\psi(w+\tfrac{1}{2}) - \log w = \int_0^\infty e^{-wt}\left(\frac{1}{t} - \frac{1}{2\sinh(t/2)}\right) dt. \tag{7}$$

For clarity, the Frullani term adds $(e^{-wt} - e^{-(w+1/2)t})/t$ to the integrand from Eq. 5.9.13, which is exactly (7). By (6), the parenthesized kernel lies between $t/24 - 7t^3/5760$ and $t/24$. Integration proves (4). Also, differentiating the standard digamma integral [2, Eq. 5.9.16] gives

$$\psi'(w+\tfrac{1}{2}) = \int_0^\infty \frac{te^{-wt}}{2\sinh(t/2)}\, dt.$$

The lower bound in (6) makes this at least $\int_0^\infty e^{-wt}(1 - t^2/24)\, dt$, which is (5). All integrals converge for $w > 0$, and differentiation is dominated on each compact positive w-interval. □

Proof of Theorem 1. Set $u = x + 1 \geq 1$, $\delta = \alpha - \frac{1}{2} \in [-\frac{1}{2}, \frac{1}{2}]$, and

$$F_u(\delta) = \log \Gamma(u) + (\tfrac{1}{2} + \delta) \log u - \log \Gamma(u + \tfrac{1}{2} + \delta) - \frac{1}{8(u+\delta)}.$$

All terms are defined because $u + \delta \geq 1/2$. We have

$$F_u'(\delta) = \log u - \psi(u + \tfrac{1}{2} + \delta) + \frac{1}{8(u+\delta)^2}, \tag{8}$$

$$F_u''(\delta) = -\psi'(u + \tfrac{1}{2} + \delta) - \frac{1}{4(u+\delta)^3} \leq -\frac{1}{u+\delta} - \frac{1}{6(u+\delta)^3} < 0. \tag{9}$$

Thus F_u is strictly concave on the full interval.

By (4), the slope at zero satisfies

$$0 < \frac{1}{12u^2} \leq F_u'(0) \leq b(u) := \frac{1}{12u^2} + \frac{7}{960u^4}. \tag{10}$$

Let

$$h = \frac{1}{10}, \qquad v = u + h, \qquad K(u) = \frac{1}{v} + \frac{1}{6v^3}.$$

For $0 \leq \delta \leq h$, (9) gives $F_u''(\delta) \leq -K(u)$. To locate the maximum, observe the exact identity

$$\frac{K(u)}{10} - b(u) = \frac{P(u)}{960u^4(10u+1)^3}, \tag{11}$$

where, writing $u = 1 + y$,

$$\begin{aligned} P(1+y) = {} & 16363 + 211070y + 759380y^2 + 1274440y^3 \\ & + 1128960y^4 + 515200y^5 + 96000y^6 > 0 \qquad (y \geq 0). \end{aligned}$$

Therefore $F_u'(h) \leq b(u) - hK(u) < 0$, while $F_u'(0) > 0$. Strict concavity places the global maximum on $[-\frac{1}{2}, \frac{1}{2}]$ inside $(0, h)$.

On $[0, h]$, integration of the second-derivative bound and completion of the square yield

$$F_u(\delta) \leq F_u(0) + F_u'(0)\delta - \tfrac{1}{2}K(u)\delta^2 \leq F_u(0) + \frac{F_u'(0)^2}{2K(u)}.$$

Lemma 1 gives $F_u(0) = c(u) - 1/(8u) < H(u) - 1/(8u)$. Using (10), the global maximum is consequently less than $-R(u)$, where

$$R(u) = \frac{1}{8u} - H(u) - \frac{b(u)^2}{2K(u)}. \tag{12}$$

It remains to check a rational inequality. Direct algebra gives

$$R(u) = \frac{N(u)}{129024000u^8(u+1)^5(2u+1)^4(300u^2+60u+53)}. \tag{13}$$

The following table gives every coefficient c_j in $N(1+y) = \sum_{j=0}^{16} c_j y^j$:

j	c_j	j	c_j
0	32048581152	9	124606427560816
1	673338924432	10	65045265578720
2	5218137436272	11	26078831078080
3	22582411828872	12	7883010992000
4	63881423270522	13	1738707609600
5	127854445158661	14	264182784000
6	189369613243226	15	24729600000
7	213134658410116	16	1075200000
8	184998793083656		

Every coefficient is positive. Hence $R(u) > 0$ for all $u \geq 1$, which proves $F_u(\delta) < 0$ on the entire required interval and establishes (1).

To prove optimality, specialize to $\alpha = 1/2$, so $I(x, 1/2) = c(u)$ and $x + \alpha = u - 1/2$. For any fixed $C > 4$, Lemma 1 gives

$$c(u) - \frac{1}{8u + C - 4} = \frac{C - 4}{64u^2} + O(u^{-3}) > 0$$

for all sufficiently large u. Such a constant C cannot give a valid upper bound. The proved inequality with $C = 4$ therefore has the largest possible additive constant. □

Exact verification and scope. The companion `verify_polynomials.py` uses only exact standard-library integer and rational arithmetic. It constructs the rational functions (11) and (12) directly from their definitions, verifies both displayed polynomial identities by formal cross-multiplication, and checks the shifted coefficients by independent binomial and Horner substitutions. Both checks concern exact polynomials, so they certify the infinite range without a numerical grid. The main identity was also derived independently with symbolic rational simplification.

The argument includes $x = 0$, $\alpha = 0$, $\alpha = 1$, and arbitrarily large x. All denominators are strictly positive. The strict midpoint bound and the positive polynomial give strictness at every finite point; sharpness of the additive constant arises at infinity. No part of the source's conjectured inequality is assumed.

Tasks 2–5. Further directions after the full solve

These are proposed tasks rather than additional solved assertions. Check current literature and public claims before attacking each one.

Task 2. Determine the infimum of the normalized deficit

$$\left(x + \alpha + \tfrac{1}{2}\right)^3 \left(\frac{1}{8(x + \alpha) + 4} - I(x, \alpha)\right)$$

on the full domain, and identify its minimizing points or limiting sequences.

Task 3. For each $x \geq 0$, determine the unique maximizer of the difference between $I(x, \alpha)$ and its upper bound. Prove monotonicity properties of this maximizer and obtain a uniformly controlled asymptotic expansion.

Task 4. Classify the parameter pairs in the larger domain $x > -1$, $0 \leq \alpha \leq 1$, $x + \alpha > -1/2$ for which the same bound remains valid.

Task 5. Find a direct integral or monotonicity proof of the sharp additive constant 4 that also proves the full bound without the degree-six and degree-sixteen polynomial certificates.

References

[1] M. Schmidlin, *On Gautschi & Stirling Identities, Asymptotics and Inequalities for the Pi (or Gamma) Function*, arXiv:2601.21906v1 (2026), https://arxiv.org/abs/2601.21906.

[2] NIST Digital Library of Mathematical Functions, Eqs. 5.9.10, 5.9.13 and 5.9.16, https://dlmf.nist.gov/5.9.